Gentle by knowing: person knowledge protects restricted joints better than force thresholds in simulated robotic arm lifting

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***Abstract*—Many care recipients have restricted shoulders or elbows, which a robot lifting their arm can press into the end of range. Contact-force limits do not sense the loaded joint when a robot supports a limb from outside. Gentleness may require knowing the person and declining. In a musculoskeletal simulation of forearm lifting, controllers know nothing (a fixed rule), a flag, joint ranges with 20° error (used by a body model or a learned policy), or everything (an oracle). A *harmful outcome* is a robot-caused end-range joint load of at least 0.5 N·m, a mechanical-exposure proxy. A *safe completion* is a completed lift below it. Hypotheses were specified in a locally kept protocol and tested on new persons. The tightest force-magnitude guard lowers mean load by 29–36%, moves harmful outcomes by 0–7 points and loses 38–51% of safe completions. With a complete record, the learned policy reduces harmful outcomes from 110 to 25 of 300 persons (safe completions 174 vs. 177, non-inferiority not established) by declining 91 and, post hoc, by harming fewer of those it lifts. At 2.5 N·m the counts are 12 harmful and 186 safe against 80 and 203. The reduction replicates under changed posture, records, population, muscle dynamics and body variation, but training and test share one body structure and joint-limit geometry. It shrinks by almost half with half of the restrictions unrecorded, and real records are sparser. Post hoc, with one learner throughout, per-joint flags give almost all of the benefit.**

***Index Terms*—Physical human-robot interaction, caregiving robots, gentle handling, safety, musculoskeletal simulation, joint contracture, range of motion, abstention**

I. Introduction

Lifting a person's arm (to wash the axilla, pass a sleeve or reposition the person) is a common care action and a target for assistive robots [1]–[4]. Doing it gently matters because many care recipients cannot move freely. About half of adults admitted with stroke develop at least one contracture within six months [5], and contractures are common in long-term care [6]. Bilateral severe spastic knee contractures independently predict minimal-trauma fractures in long-term care residents [7]. In people with severe motor and intellectual disabilities, some fractures occur during diaper changes, repositioning or transfers [8]. In hemiplegia, shoulder pain is associated with overhead-pulley exercise [9] and with needing transfer help [10].

Collaborative-robot safety bounds contact force and pressure with limits derived from pain-onset thresholds [11], and injury knowledge enters robot control for impacts [12]. Adjacent standards go further: wearable assistive robots are designed to apply assist force within the user's passive range of motion [13], rehabilitation-robot requirements address movement beyond pre-set, patient-specific limits [14], [15], and personal-care robots seek external assistance or stop when decision uncertainty remains unacceptable [16]. None of these states what a robot that moves a limb by end-effector contact must know about the person, or whether sensing contact force is enough.

Our key insight is that, when a robot supports a multi-joint limb from outside, the joint it loads is not on the axis of its force sensor. A harmful joint load often adds only a few newtons at the contact point, hidden under the limb's own weight, so a force guard trips only after the joint is loaded (Figure 1C). A force-magnitude guard alone therefore cannot keep a lift gentle. It must know the person before contact and be willing not to lift. Human carers work this way: they read in the chart that "the left shoulder is stiff" and handle the arm differently or ask for help.

We test two hypotheses with a simulated robot that lifts the forearm of persons with restricted shoulders and elbows (Figure 1): force-magnitude guards do not reduce harmful outcomes by more than 5 percentage points, and knowledge of the person, with the option to decline, reduces them. We also test whether safe completions stay non-inferior.

Related work

Robots that move limbs. Simulation frameworks model care recipients with physical limitations [17], [18] and train assistive policies for simulated impairments [19]. Physical systems lift a bed-bound person's limb while bounding tool contact force [2], reposition limbs within age-group joint limits [3], plan dressing within each participant's measured range [20] and predict functional range from clinical scores [21]. None measures harm across levels of information about the person or compares that information with force sensing.

Force limits and joints. The same external force can correspond to very different joint loads [22]. A robot's force-torque sensor senses the rendered end-range resistance of a haptic shoulder device [23]. Industrial force limits do not transfer directly to patients, although in exoskeletons force control helps keep joints in range [24]. In a joint-aligned ankle stretcher, a torque threshold safely ends the stretch at a set peak resistance [25]. Force sensing thus works when the sensor sits

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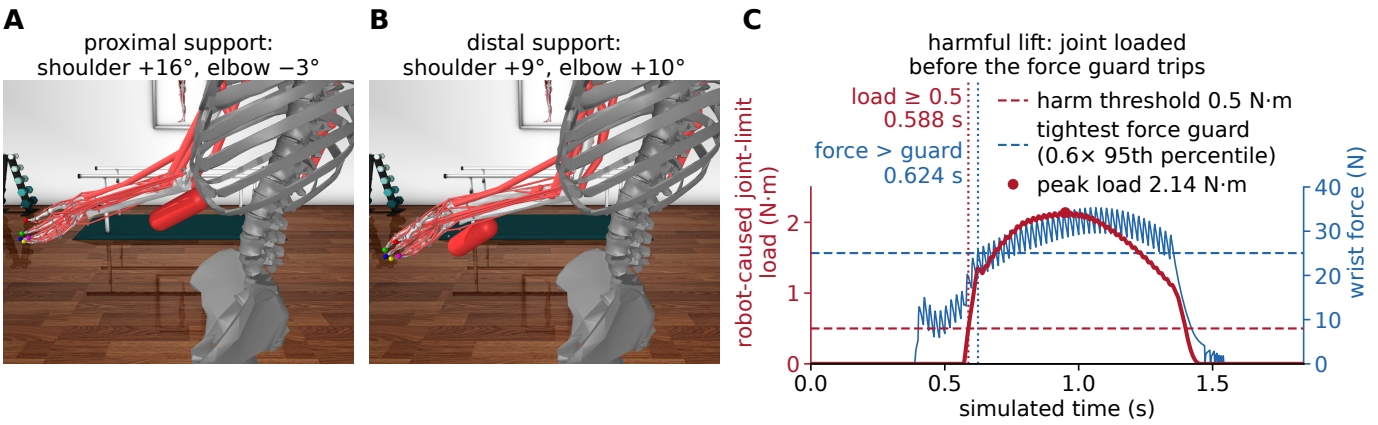


Fig. 1. The robot loads the joint before its force sensor notices. **(A, B)** A capsule supports the forearm from below and raises it 5 cm. Proximal support elevates mainly the shoulder. Distal support also flexes the elbow (joint excursions for one unrestricted simulated person, seed 104), so for a person with a restricted shoulder or elbow the wrong site presses that joint into the end of its range. **(C)** One harmful lift of the fixed rule (seed 2222, restricted elbow): the robot-caused joint-limit load passes 0.5 N·m at 0.588 s, before the wrist force exceeds the tightest force-guard threshold (25.05 N, 0.6 times the 95th percentile of harmless force) at 0.624 s.

on the joint's axis, but in end-effector support of a multi-joint limb a force-magnitude threshold falls short.

Value of information. Declining is the reject option of classification [26], and we measure an empirical analogue of the value of information [27]. Comparing a baseline, an inference model and an omniscient planner has precedent in autonomous driving [28], and assistive robotics compares generic and personalised controllers, interaction models and preference models [29]–[31]. To our knowledge, no work varies the person-specific body information available to a physically assistive robot and reports harm and safe completion at each level, with declining as an option.

Contributions

1) Evidence that tight force-magnitude guards lower the load but do not keep it under 0.5 N·m, so they do not substitute for knowledge of the person (Figure 2).
2) An *information ladder* (Figure 3), in which a learned policy cuts harmful outcomes to about a quarter of a fixed rule's with a complete record and loses much of this when the record misses restrictions (Figure 4).
3) A post hoc analysis with one learner at every level, showing that the robot needs to know which joint is restricted, not by how many degrees, and an account of what does not help, with every hypothesis reported (Supplementary Table S1).

II. Materials and Methods

A. Simulated body and population

The simulation uses the myoArm model of the MyoSuite framework [32] (version 2.12.2) in MuJoCo 3.6 [33], trunk upright, with the humerus free of the thorax so that the arm hangs from the shoulder complex. A stretch reflex with muscle velocity and length feedback drives the muscles, following Yu et al. [34], with a 25-ms afferent delay and a tonic baseline activation of 0.01 (our choices, whereas Yu et al. use 30 ms). The simulated person neither assists nor resists voluntarily. Muscle strength, passive muscle force and joint damping vary log-normally across persons ($\sigma = 0.25$, segment masses and inertias $\sigma = 0.175$, Supplementary Table S3).

Each person independently has a restricted shoulder elevation with probability 0.4 and a restricted elbow flexion with probability 0.3, so 58% have at least one restricted joint, similar to 52% (any joint) after stroke [5] and 61% (shoulder 44%, elbow 21%) in one nursing-home sample [6]. The per-joint rates exceed those after stroke (25% shoulder, 22% elbow) and, with the small free ranges, represent a severely affected group. A restricted joint moves a free range drawn from $U(4°, 16°)$ (shoulder) or $U(0°, 10°)$ (elbow) above the person's settled posture before its limit engages. As stated in the protocol, the free ranges overlap the task's joint excursions (roughly 7°–16° at the shoulder and up to 10° at the elbow, Figure 1), so absolute harm rates reflect the population design and only paired differences between controllers are interpreted. A *shifted* population has restriction probabilities 0.6 (shoulder, $U(2°, 12°)$) and 0.5 (elbow, $U(0°, 8°)$). The joint limit is hard by default. Under *graded end-range resistance*, resistance ramps up over a window of $w \in \{6°, 12°, 20°, 25°\}$ centred on the limit (derived by us from torque-angle data reporting no window: [35]–[37]).

B. Robot and task

The end effector is a position-servoed capsule with three translational and three rotational joints (effectively five degrees of freedom, because spin about the capsule's own axis changes nothing, 6,000 N/m, 20 N·m/rad, 50 Hz). It approaches the forearm from below, raises the forearm centre by 5 cm in closed loop on the observed forearm height, and holds for 0.5 s. A *strategy* is a support site (8 sites, 3 cm apart), an approach angle ($-40°$, $-20°$, $0°$) and a speed (0.10 or 0.25 m/s): 48 strategies. No single strategy is safe for everyone (Figure 1).

C. Gentleness: harmful outcomes and safe completions

Harm is the load the robot presses into the shoulder-elevation and elbow-flexion limits: the peak 50-ms moving average of the joint-limit constraint torque (excluding the first 100 ms), summed over the two joints. From round 23 onward, limit torque present before contact is subtracted. With a hard limit this changes at most 2 of 300 classifications under F, and running 12° this way on other persons changed no conclusion (Supplementary Table S2). An outcome is *harmful* if harm ≥ 0.5 N·m (primary threshold). We also report 2.5 N·m, near the mean minimum torque (2.7 ± 1.1 N·m) at which therapists reached passive elbow end range in children after fracture [38], a threshold sweep and a threshold-free mean load. This anchor is pediatric. Adult passive-torque studies use higher values (4 and 8 N·m in shoulder rotation, [35], [36], 10 N·m at the stroke elbow, [37]), so we read 2.5 N·m as one point on the sweep, which reaches 5 N·m, below the two higher values. Such loads are routine in supervised assessment. The thresholds therefore mark unintended, unmonitored end-range loading by a robot (a mechanical-exposure proxy, not injury). A *safe completion* is a completed lift with harm below the threshold. A declined lift counts as neither. No controller except the oracle sees harm.

D. What the robot may know

The *care record* gives each joint's free range with Gaussian error of SD $\varepsilon \in \{4°, 10°, 20°\}$. Standard errors of measurement of 14–25° between raters for symptomatic shoulders [39] and 6–10° on retest for the elbow and wrist after stroke [40] make 20° the realistic default. Unrestricted joints are recorded as unrestricted, so a *complete* record always tells which joints are restricted (an idealisation). Real records are often categorical, such as the Minimum Data Set 3.0 item that codes shoulder, elbow, wrist and hand together [41], and incomplete: in one nursing-home study, reduced range was found in 69% of residents but documented for 4% [42]. We also use a categorical flag (shoulder and elbow merged) and a *half-documented* record that omits each restricted person with probability 0.5. The robot observes the settled shoulder and elbow angles before contact and its wrist force during the lift.

E. Controllers

Controllers differ in what they know, whether they may decline and how they decide. Section III-D holds the decision rule fixed.

- **F (fixed rule).** Among strategies completing $\geq 90\%$ of 60 training persons, the one with the lowest median harm.
- **R (force-magnitude guard).** Runs F. If the wrist-force magnitude exceeds m times the 95th percentile of harmless training lifts ($m \in \{0.6, 0.8, 1.0, 1.2\}$), it stops, retracts and tries the best strategy supporting at least 9 cm away and then the next-best strategy (up to three attempts).
- **CAT, CAT_{half}.** Decline if the complete or half-documented flag marks a restriction, otherwise F.
- **S (record threshold).** Decline if a recorded free range is below 3.5° (locked on 280 earlier persons), otherwise F.
- **M (population body model).** A library of 199 simulated unrestricted bodies predicts each joint's excursion and each strategy's completion. M keeps F if F is predicted to stay 3° inside both recorded ranges. Otherwise it uses the margin-keeping strategy with the highest predicted completion (≥ 0.8). Otherwise it executes the least-overshooting strategy if the overshoot is at most 3°, and otherwise declines.
- **L (policy learned offline in simulation).** From 400 simulated training persons with outcomes for all 48 strategies, a k-nearest-neighbour model ($K = 25$) over six standardised features (two recorded free ranges clipped at 30°, two unrestricted flags and two settled joint angles) estimates each strategy's probability of a safe completion. L picks the best strategy and declines if its estimate is below $\tau = 0.65$: the training-set value with the fewest harmful outcomes among those keeping safe completions within 2.5 percentage points of F, selected with each training person among its own neighbours (Supplementary Table S2 gives a leave-one-out re-selection). Persons recorded as unrestricted receive F. A four-feature variant without posture ($\tau = 0.60$) was specified in advance for round 25.
- **O (oracle).** Knows every outcome. Picks a safe completion if one exists, otherwise declines.

F. Protocol and statistics

The protocol document began as the plan of an earlier phase (rounds 1–16, other tasks, main hypotheses not supported). Rounds 17–26 (Supplementary Table S6, with 18R the frozen replication of round 18) were added as dated amendments, each written before that round's data were generated. It was kept locally, not externally registered, so no outside record dates the amendments. Every confirmation set consists of simulated persons never used before. Earlier confirmation persons were later reused for development and training.

All comparisons are paired. Differences in harmful-outcome and safe-completion rates carry 95% confidence intervals (paired t). The prespecified analyses use $t = 2.1$ for $n > 20$, slightly conservative, and post hoc analyses the exact t quantile, except the guard re-analysis (Supplementary Table S7). A reduction counts if the upper bound is below zero. For force guards the protocol required an upper bound below -5 percentage points, which cannot support the hypothesis of no large reduction. Post hoc, we therefore test non-superiority: the lower bound must stay above -5 points, and a cell that fails is inconclusive. Non-inferiority uses margins fixed in advance (2–2.5 points, or a stated count), equivalence ± 0.05. Differences are proportions (-0.19 = 19 percentage points fewer). Rounds 17–26 are not corrected for multiplicity, which limits their confirmatory weight. Post hoc analyses tune baselines on the 400 training persons only and evaluate them on the 600 persons of the ladder set and replication (Supplementary Table S6), with complete outcome grids. Supplementary Table S2 lists all deviations. Code was written and run with Claude Opus 5.5 (Anthropic) under the author's direction, and the author verified it.

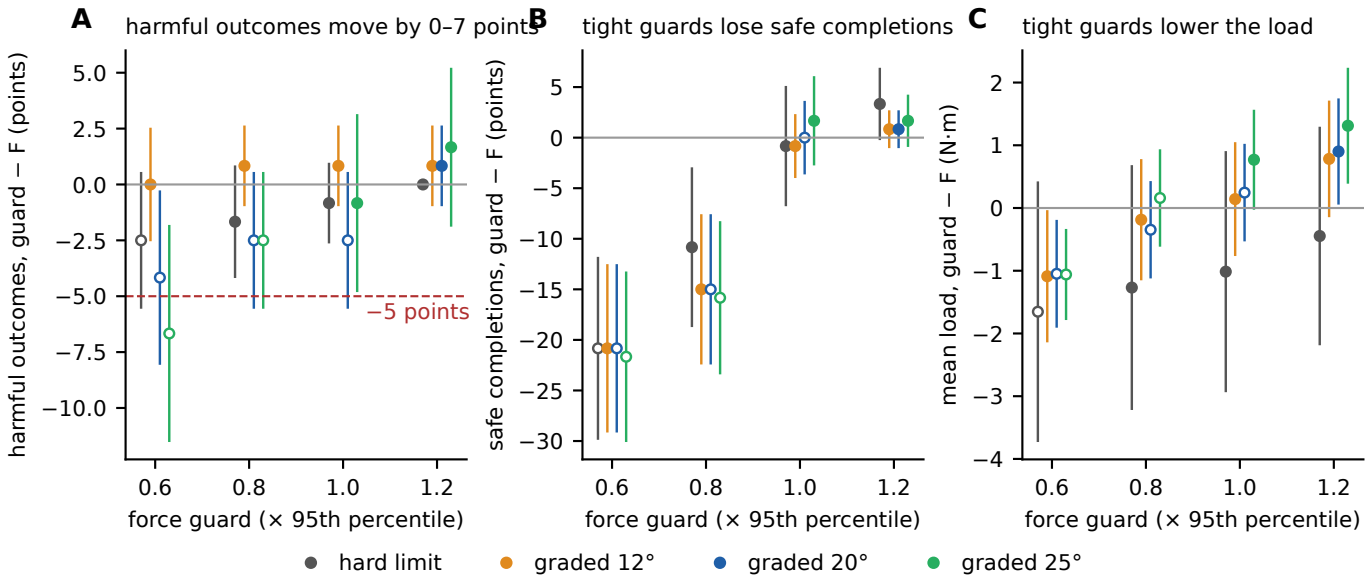


Fig. 2. Tight force-magnitude guards lower the load and lose safe completions, while harmful outcomes change by at most 7 points. Paired differences from the fixed rule F (95% confidence intervals) in **(A)** harmful-outcome rate, **(B)** safe-completion rate and **(C)** mean load, for four guard thresholds under a hard limit (120 new persons) and graded end-range resistance of 12, 20 and 25° (120 other new persons, the same under each width). Dashed line: −5 points. Hollow markers: the interval in **(A)** reaches past −5 points, so non-superiority (post hoc test) is inconclusive. The 6° cells are in Supplementary Table S7.

III. Results

A. Force-magnitude guards do not substitute for knowledge

The fixed rule F harms 34–40% of persons in every standard-population hard-limit set of at least 120 persons (e.g. 229 of 600). A guard stopping a lift past 0.5 N·m still scores a harmful outcome, so only the mean load shows what it removes. In the conditions of Figure 2 the tightest guard lowers mean load by 29–36% (hard limit 5.43 to 3.77 N·m, interval including zero, 25° 2.92 to 1.86), leaves 36–48% of persons above 0.5 N·m and loses 38–51% of safe completions. It moves the harmful-outcome rate by 0 to 7 points, with upper bounds below zero at 20 and 25°. Post hoc, non-superiority to 5 points is shown in 10 of these 16 cells and is inconclusive in 6 (12 of the 32 cells of Supplementary Table S7), including the tightest guard at 25° (−7 [−11, −2]). Looser guards do not lower the load reliably, and at 25° the loosest raises it (4.23 vs. 2.92 N·m) through its retries: first attempts average 2.63 N·m, the 33 second attempts 10.91.

Between 95% and 100% of harmful outcomes occur during the first attempt. Post hoc, each guard prevents only 0–13% of F's harmful outcomes (0–10% with a hard limit), mostly by aborting the lift. In the remaining harmful first attempts the guard never trips in 33–79% (hard limit) and 42–95% (graded resistance) of cases, and when it trips, the joint load has already reached 0.5 N·m. Among the 600 persons of the ladder set and replication, every harmful lift of F stays below the ISO/TS 15066 quasi-static force limit for the lower arm (160 N, post hoc).

B. The information ladder

Knowledge helps mainly by telling the robot whom not to lift. The body-model rule M reduces the harmful-outcome rate by 14–26 percentage points in every standard-population condition with a complete record (e.g. −0.19 [−0.23, −0.16], 113 vs. 229 of 600, safe completions 351 vs. 344), by 9 points with half of the restrictions unrecorded and by 37 in the shifted population. Without its decline option it is not shown to reduce harmful outcomes (Section III-E).

Figure 3 shows the ladder set of 300 new persons, including those declined, under the controllers and under one learner at six levels of knowledge. Declining every flagged person removes all harmful outcomes but 37% of safe completions (112 safe, 0 harmful). With half the flags missing it leaves 60 harmful outcomes. Ranges with 20° error plus the record threshold S or the body model M reduce harmful outcomes by 39–54%. The learned policy L reduces harmful outcomes to 25, fewer than M (−0.14 [−0.19, −0.09]) with no fewer safe completions than M (+0.04 [0.00, +0.08]). Against F it completes 174 vs. 177 lifts safely, and prespecified non-inferiority is not established (−0.01 [−0.05, +0.03]). Compared with declining every flagged person, L completes 62 more lifts safely at the cost of 25 harmful outcomes, so with a decline counted as free L is preferable while a harmful outcome weighs less than about 2.5 completions. Record accuracy matters less once L is used (4°: 184 safe, 20 harmful).

Mean robot-caused end-range load (declines as zero) is 6.73 N·m under F, 3.19 under M and 0.79 under L (L − F: −5.94 [−7.78, −4.10]). The advantage is not only lifting fewer people than F: post hoc, on the 423 of 600 persons L lifts, L harms 53 and F would harm 95 (−0.10 [−0.13, −0.07]).

C. Robustness of the learned policy

On new persons (the replication, Figure 4, Supplementary Table S5), L replicates its reduction relative to F (−0.27

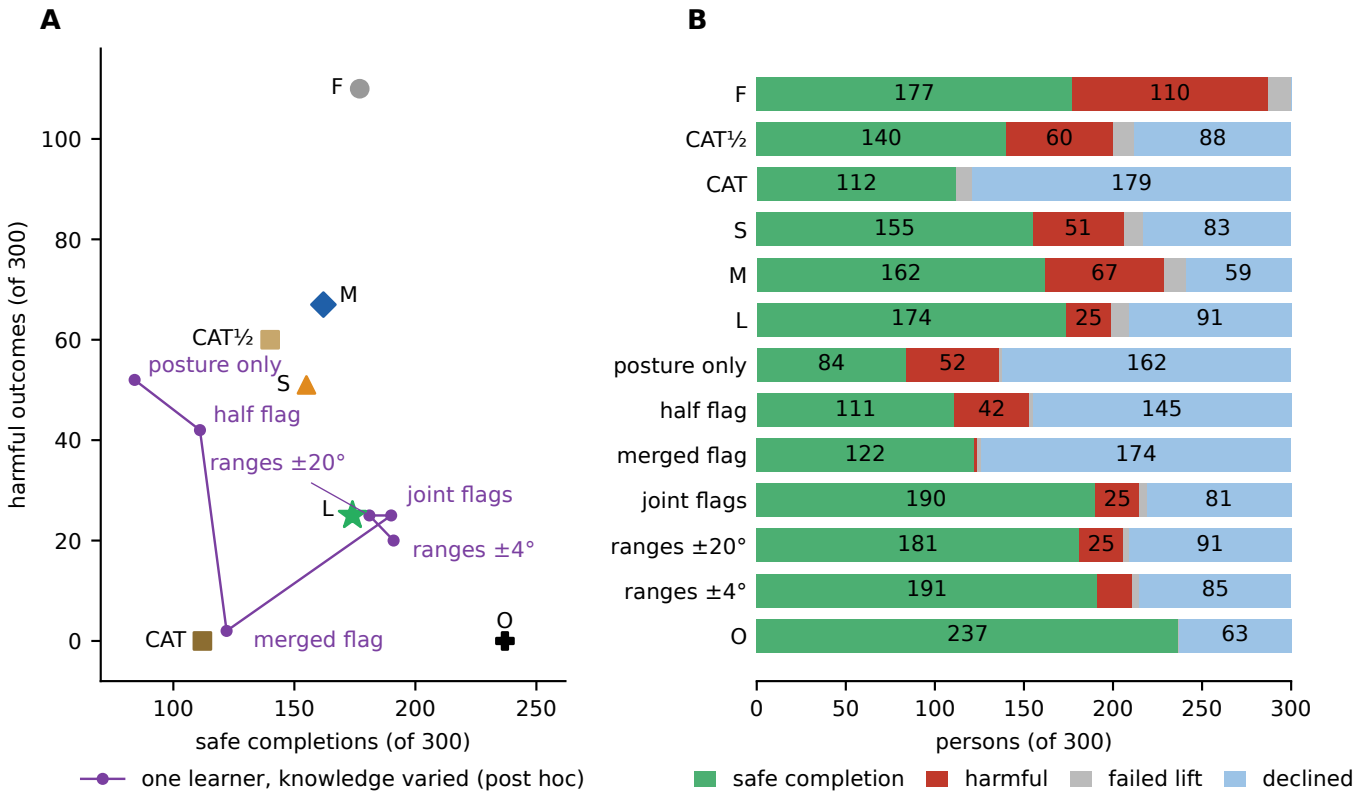


Fig. 3. Knowing the person reduces harm, mostly by deciding whom not to lift. Ladder set, 300 new persons, harmful ≥ 0.5 N·m. **(A)** Harmful outcomes against safe completions (lower right is better). Markers are the prespecified controllers: fixed rule F, flag rules CAT_{half} and CAT, and S, M and L with ranges at 20° error, and the oracle O. The line joins one learner with the rule held fixed at six levels of knowledge (post hoc, $\tau = 0.65$). **(B)** What happens to all 300 persons under each controller, declines included. Counts are in Supplementary Tables S4 and S8.

[−0.32, −0.21]) and to M (−0.08 [−0.12, −0.03]), but its safe-completion non-inferiority to M does not replicate (−0.01 [−0.05, +0.03]). Relative to M, the reduction holds with Gaussian posture noise (SD 5°) or without posture input. Relative to F and M, it holds in the shifted population (170 vs. 38 harmful of 250, without retraining) and under 20° graded resistance. L's safe-completion rate relative to F is −0.01 [−0.06, +0.04] in the replication, +0.02 [−0.03, +0.07] in the shifted population and +0.06 [0.00, +0.12] under 20° graded resistance.

Record completeness matters more than record accuracy. With half of the restrictions unrecorded, L's reduction relative to F shrinks from −0.27 to −0.15 (64 vs. 108 harmful), and CAT_{half} harms fewer persons than L (50 vs. 64, +0.05 [+0.02, +0.07]) at 29 fewer safe completions. Unrecorded persons are harmed at exactly the fixed-rule rate (50 vs. 50 of 85). With documentation as sparse as reported [42], little benefit would remain without a screen before robot use.

With threefold stretch-reflex gains or individual variation widened from 0.25 to $\sigma = 0.40$, L's reduction relative to F and M holds, with safe completions 3.3 and 0.3 points below F's. These perturbations keep the joint-limit geometry, and F's harm rate barely moves (36% and 40%), so they test muscle dynamics and body variation, not body structure.

In a post hoc threshold sweep (0.25–5 N·m, Supplementary Figure S1), L has the fewest harmful outcomes among F, S, M and L at every threshold, but F completes more lifts safely than L at almost every threshold. At the prespecified secondary 2.5 N·m, L completes 4–8 points fewer lifts than F (6–8 with a complete record).

D. What the robot needs to know (post hoc)

The controllers of Figure 3 change rule as well as knowledge. To hold the rule fixed, one learner (L's method, $\tau = 0.65$, deciding for every person) runs on each level below the oracle, with the settled posture always included (Supplementary Table S8). On the 600 persons it gives 169 safe and 96 harmful with posture alone, 239 and 4 with a merged flag (348 declined), 366 and 46 with per-joint flags and 354 and 53 with ranges at 20° error (F: 348 and 218). At no more than 46 harmful outcomes along the τ sweep (descriptive, read on these persons), safe completions are 41, 280, 366 and 335: a merged flag adds 239 to posture and per-joint flags 86 more, so the gap between CAT and L is one of knowledge as well as rule. With 20° error, recorded degrees are not shown to add to the flags (harmful +0.01 [−0.01, +0.03], safe −0.02 [−0.04, 0.00]).

E. What does not help

Choosing the support site without declining is not shown to reduce harmful outcomes (−0.03 [−0.09, +0.02], prespecified null). Identifying the body by two gentle probe lifts is supported in a first set ($n = 40$) but not in its replication (120

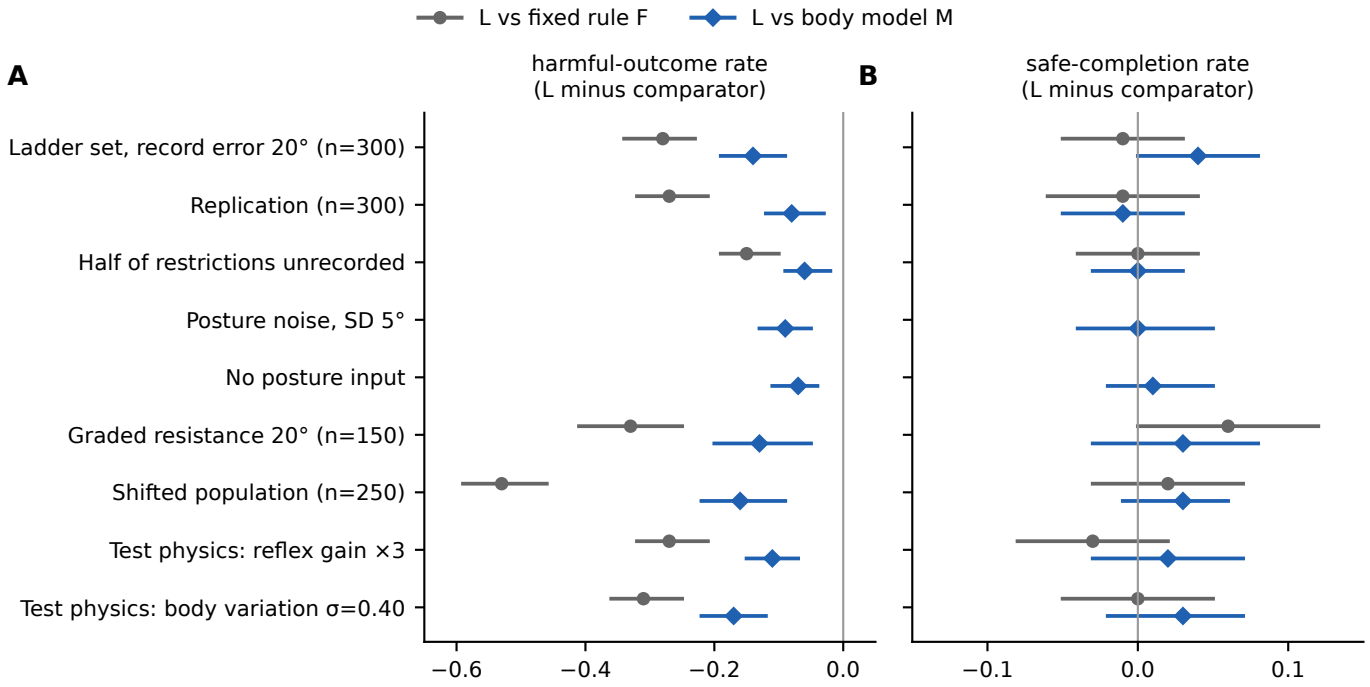


Fig. 4. The learned policy's reduction in harm holds in every prespecified confirmation condition but shrinks when half of the restrictions are unrecorded. Paired differences (95% confidence intervals) in **(A)** harmful-outcome rate and **(B)** safe-completion rate between L and the fixed rule F (grey) or the body-model rule M (blue), harmful $\geq$ 0.5 N·m. The nine rows come from five independent person sets. L, M and F are not retrained. The no-posture variant is the separately specified four-feature policy. Posture noise is Gaussian with SD 5°. Posture conditions are compared with M only, as prespecified. Counts are in Supplementary Tables S4 (ladder set) and S5.

persons). In a later set the probes load a restricted joint beyond 0.5 N·m in 18 of 71 persons, and probing harms more persons than the same rule without it (29 vs. 21 of 120).

IV. Discussion

Why knowledge, and not force

Over the 600 persons and eight support sites (post hoc), 0.5 N·m corresponds to 0.8–4.8 N at the contact with the full lever arm. Along the lift direction it is 1.6–6.8 N at the elbow and a median of 8 N at the shoulder (5th to 95th percentile 5–109 N, because the lift line can pass close to the shoulder). Most of these are small against a forearm-and-hand weight of about 17 N and guard thresholds of 25–50 N. At the adult torques cited in Methods these forces are 8–20 times larger, so this argument holds at low thresholds. A force-magnitude guard therefore trips late, if at all, and can lower the load without keeping it under 0.5 N·m. Knowledge acts before contact, where the decision that matters most is whether to lift at all. Force limits remain necessary for contact safety but are not sufficient for joints. Declining echoes robots that ask for help under ambiguity [43], here triggered by predicted harm to the body.

What kind of knowledge

Here the robot needs no detailed personal model: probing adds harm, and recorded degrees add little to per-joint flags (post hoc). What matters is knowing *which* joint is restricted, together with a policy learned from the simulated outcomes of many similar persons.

The cost of declining

Declining hands the task to a human carer, whose handling carries its own risk (harm counts here are robot-caused). It mirrors the care rule to stop at resistance and report. L declines 30% of persons, the oracle 21%. Because τ keeps safe completions close to F's, similar completion counts are partly a design choice (Supplementary Figures S1 and S2). A deployed system would set τ with the care team. The simulated persons do not wince or speak. The results apply most directly to people who cannot give such cues, such as those with advanced dementia or reduced consciousness.

Implications for design

Robots that move limbs by end-effector contact should (i) obtain joint-level information before contact (at least a per-joint, passive pass/fail screen for the range the task needs, repeated after significant change, which current merged-limb functional items do not provide), (ii) include an explicit option to decline and hand over the task, and (iii) not rely on force-magnitude thresholds alone to protect joints. Our results assume a complete record, which care settings rarely have, so the screen must reach every person before first use. The first two echo adjacent standards [13], [14], [16].

Limitations and future work

All results are simulated, and training and test simulators share the musculoskeletal structure and joint-limit geometry. An arm phantom with adjustable limits and a change of body

geometry between training and test are also needed. The harm measure has no dose-response and depends on the simulator's constraint model. The model restricts elbow flexion, not the clinically more common loss of extension, assumes independent restrictions with small uniform free ranges, and idealises the complete flag. The study covers one task, single-point support, a passive recipient and only force-magnitude guards with thresholds set across persons. Guards and L never ran on the same persons. The most informative next comparison is a guard that estimates joint torque from the force vector with a limb model. Person-relative thresholds, detecting a stalled lift, compliant control, slower lifting and two-point support, as carers use, may also narrow the gap. The protocol was kept locally, and the analyses of what the robot needs to know are post hoc.

Conclusion

In simulated arm lifting, gentleness depends less on sensing the contact than on knowing, before contact, which joint is restricted and when not to lift. Tight force-magnitude guards lower the load but do not keep it under 0.5 N·m, and unrecorded restrictions remain unprotected. With a complete record, a learned policy has about a quarter of a fixed rule's harmful outcomes, but its safe-completion non-inferiority is not established, and training and test share the simulator's body structure and joint-limit geometry, so transfer to other bodies is untested.

Conflict of Interest Statement

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI Statement

The author declares that generative AI was used in the creation of this manuscript: Claude (model Claude Opus 5.5, Anthropic, accessed through Claude Code between August and November 2026) was used to write and run simulation and analysis code, search and summarise literature, check numbers and references, and draft and edit the text, under the author's direction. The author reviewed and verified all code, results, references and text and takes full responsibility for the content.

Author Contributions

M-HW: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing (original draft), Writing (review and editing).

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Supplementary Material

Supplementary Table S1 lists every prespecified hypothesis with its outcome, Table S2 deviations and reuse of persons, Table S3 simulation parameters, Tables S4 and S5 the counts behind Figures 3 and 4, Table S6 the confirmation rounds, Table S7 the force-guard cells and Table S8 every controller by knowledge and rule. Supplementary Figure S1 shows the post hoc harm-threshold sweep and Figure S2 the post hoc trade-off between safe completions and harmful outcomes.

Data Availability Statement

The simulation code, the dated protocol with all amendments, per-episode outcome data for every confirmation set, and scripts that regenerate every number, table and figure in this article are available at https://github.com/maurice1128/gentle-by-knowing.

Ethics Statement

This study used only computer simulation. No human participants, human data or animals were involved, and ethical approval was not required.

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